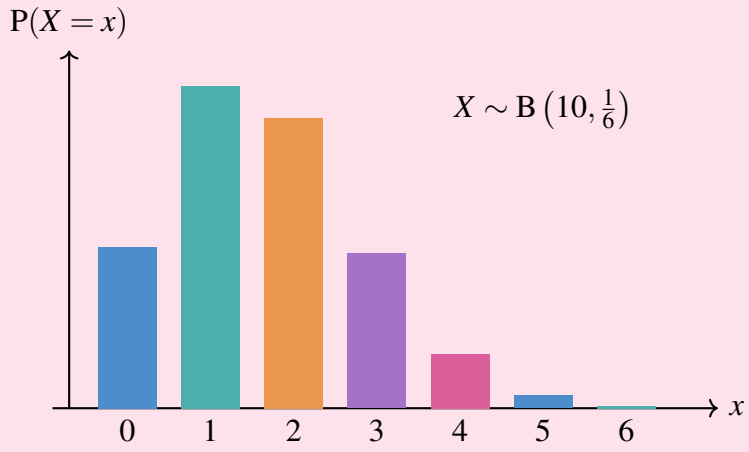
Binomial Distribution

IB Maths AA SL/HL 4.10 · Notes · Questions · Worked Answers

1. Recognising binomial & finding probabilities
2. Mean, variance & expected numbers
3. Finding n or p
4. Conditional probability & multi-step problems
5. Extended problems



No calculator Calculator (TI-Nspire CX II) allowed

Every item in the Index is a clickable link. Each topic has its own colour.

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= no calculator = calculator allowed

Sources: **RV** = Revision Village practice questions (Practice Set 2 numbering); **SME** = Save My Exams IB *Binomial Distribution* (Medium / Hard / Very Hard); **Oxford** = Oxford IB textbook exercises 14D, 14E, 14G and exam-style questions (Practice Set 1); **Y13 Worksheet** = IB Year 13 Binomial & Normal worksheet; **Kognity** = Kognity practice sets (B, C, D). These sources give no exam years. Repeated questions appear only once.

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Key Notes

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1. When is X binomial? Check "BINS"

Binary	each trial has only two outcomes: success or failure
Independent	the trials do not affect each other (e.g. *with* replacement)
Number	the number of trials n is fixed
Same	the probability of success p is the same on every trial

Then $X \sim \mathrm{B}(n, p)$, where X = number of successes in n trials. A "success" is whatever you are counting (it can be a faulty item).

Not binomial: counting until the first success (no fixed n); without replacement from a small bag (p changes).

2. Formulae (in the IB formula booklet)

Probability of exactly r successes	$\mathrm{P}(X = r) = \binom{n}{r} p^r (1-p)^{n-r}, \quad r = 0, 1, \ldots, n$
Mean (expected value)	$\mathrm{E}(X) = np$
Variance	$\mathrm{Var}(X) = np(1-p)$
Standard deviation	$\sigma = \sqrt{np(1-p)}$

Non-calculator: $\mathrm{P}(X \geq 1) = 1 - \mathrm{P}(X = 0) = 1 - (1-p)^n$. Give exact answers as fractions when asked.

3. Using the TI-Nspire CX II

menu → **5: Probability** → **5: Distributions** →

- **D: Binomial Pdf** — one value: $\mathrm{P}(X = r)$ = `binomPdf(n,p,r)`
- **E: Binomial Cdf** — a range: $\mathrm{P}(a \leq X \leq b)$ = `binomCdf(n,p,a,b)` (both ends **included**)

Always write the distribution, e.g. "$X \sim \mathrm{B}(20, 0.25)$", and the probability statement before the answer.

4. Translating the words (X takes whole-number values only)

Words	**Inequality**	**GDC** ($n = 20$)
at least 4	$X \geq 4$	`binomCdf(20,p,4,20)`
more than 4	$X > 4 \Leftrightarrow X \geq 5$	`binomCdf(20,p,5,20)`
at most 4 / no more than 4	$X \leq 4$	`binomCdf(20,p,0,4)`
fewer / less than 4	$X < 4 \Leftrightarrow X \leq 3$	`binomCdf(20,p,0,3)`
at least one	$X \geq 1$	$1 - \mathrm{P}(X = 0)$

★ **Discrete vs continuous:** for the binomial, "more than 80" means $X \geq 81$. For the *normal* distribution (continuous) the bound stays 80.

5. Expected value — do not round

$\mathrm{E}(X) = np$ is a long-run average, so it need not be a whole number.

★ **Leave expected values as decimals**, even when counting people, flights or items: $138 \times 0.17 = \mathbf{23.46}$, $341 \times 0.42 = \mathbf{143.22}$.
Round to a whole number *only* if the question tells you to.

Expected number from a two-stage model: e.g. P(tall) from a normal distribution, then $n \times p$.

6. Conditional probability

$$\mathrm{P}(A \,|\, B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} \qquad \text{e.g.} \quad \mathrm{P}(X \leq 4 \,|\, X \geq 2) = \frac{\mathrm{P}(2 \leq X \leq 4)}{\mathrm{P}(X \geq 2)}$$

Top: the overlap of the two conditions. Bottom: the condition after "given that".

7. "The kth success happens on the nth trial"

Exactly $k-1$ successes in the first $n-1$ trials, **then** a success on trial n:

$$\mathrm{P} = \mathrm{P}(Y = k-1) \times p, \qquad Y \sim \mathrm{B}(n-1, p)$$

e.g. third prize on the tenth game: $\mathrm{P}(T = 2) \times p$ with $T \sim \mathrm{B}(9, p)$.

8. Finding the least n, or an unknown p

$\mathrm{P}(\text{at least one}) > 0.95 \iff 1-(1-p)^n > 0.95 \iff (1-p)^n < 0.05$.
Use logs ($n > \frac{\ln 0.05}{\ln(1-p)}$, then round **up**) or the GDC table, and give the smallest whole n.
Given $\mathrm{E}(X)$ and $\mathrm{Var}(X)$: $np = \mathrm{E}$, $np(1-p) = \mathrm{Var} \Rightarrow 1-p = \dfrac{\mathrm{Var}}{\mathrm{E}}$.
Given $\mathrm{Var}(X)$ only: solve the quadratic $np(1-p) = \mathrm{Var}$ — there may be two values of p.

9. "Within one standard deviation of the mean"

$\mathrm{P}(\mu - \sigma < X < \mu + \sigma)$: work out the end values, then round **inwards** to whole numbers.
e.g. $\mu = 240$, $\sigma = 9.80$: $\mathrm{P}(230.2 < T < 249.8) = \mathrm{P}(231 \le T \le 249)$.

10. Worked example

A fair die is rolled 10 times. X = number of sixes, so $X \sim \mathrm{B}\left(10, \frac{1}{6}\right)$.

- $\mathrm{E}(X) = 10 \times \frac{1}{6} = \frac{5}{3} \approx 1.67$, $\quad \mathrm{Var}(X) = 10 \times \frac{1}{6} \times \frac{5}{6} = \frac{25}{18} \approx 1.39$
- $\mathrm{P}(X = 2) = \binom{10}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^8 \approx 0.291$, $\quad \mathrm{P}(X \ge 1) = 1 - \left(\frac{5}{6}\right)^{10} \approx 0.838$

1 answer → No calculator **SME Medium · Q1**

[Maximum mark: 8]

(a) State the conditions that must be satisfied to be able to model a random variable X with a binomial distribution $\mathrm{B}(n, p)$. [4]

(b) A fair spinner has 8 sectors labelled with the numbers 1 through 8. For each of the following cases, state with a reason whether or not a binomial distribution would be appropriate for modelling the specified random variable.

(i) The random variable S is the number of the sector that the spinner lands on when it is spun.

(ii) The random variable W is the number of times the spinner is spun until it lands on '7' for the first time.

(iii) The random variable Y is the number of times the spinner lands on a prime number when it is spun twelve times.

(iv) On the first spin, it is a 'win' if the spinner lands on an even number. On subsequent spins it is a 'win' if the spinner lands either on the same number as the previous spin or on a factor of the number from the previous spin. The random variable L is the number of wins when the spinner is spun ten times. [4]

1 ← question *SME Medium · Q1*

(a) fixed number of trials n; each trial has two outcomes (success/failure); constant probability p; trials independent **A4**

(b)(i) **No** — S is the number shown, not a count of successes.

(ii) **No** — the number of trials is not fixed.

(iii) **Yes** — $n = 12$ fixed, success = prime ($p = \frac{4}{8} = \frac{1}{2}$), independent: $Y \sim \mathrm{B}(12, \frac{1}{2})$.

(iv) **No** — the probability of a win changes and depends on the previous spin (not independent). **A4**

2 answer → Calculator **Kognity C · Q3**

[Maximum mark: 1]

For a manufacturing company, the probability of making a faulty product is 0.02. The discrete random variable X represents the number of defective products in a batch of 20 consecutive products that the production line makes. Which of the following options is a true statement?

1. X is not binomially distributed because there is no fixed number of trials.

2. X is binomially distributed.

3. X is not binomially distributed because the probability of success is not constant in each trial.

4. X is not binomially distributed because X represents the number of failures (faulty products) and not the number of successes.

3 answer → Calculator **Y13 Worksheet · Q1**

[Maximum mark: 2]

A fair die is rolled 15 times. Let X be the number of times a six appears. Find $\mathrm{P}(X = 3)$.

4 answer → Calculator **Y13 Worksheet · Q2**

[Maximum mark: 2]

A biased coin has probability 0.6 of showing heads. The coin is tossed 10 times. Find $\mathrm{P}(X \geq 7)$, where X is the number of heads.

2 ← question *Kognity C · Q3*

Option 2. $n = 20$ fixed, two outcomes, $p = 0.02$ constant, independent: $X \sim \mathrm{B}(20, 0.02)$. A "success" can be any outcome we count — here a faulty product. **A1**

3 ← question *Y13 Worksheet · Q1*

$X \sim \mathrm{B}\left(15, \frac{1}{6}\right)$ **M1**

$\mathrm{P}(X = 3) = \boxed{0.236}$ (0.236256) **A1**

TI-Nspire CX: Question 3

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(15,1/6,3)`

4 ← question *Y13 Worksheet · Q2*

$X \sim \mathrm{B}(10, 0.6)$; $\mathrm{P}(X \geq 7) = \boxed{0.382}$ **M1 A1**

TI-Nspire CX: Question 4

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(10,0.6,7,10)`

5 answer → Calculator **Y13 Worksheet · Q3**

[Maximum mark: 2]

The probability of a student passing a test is 0.8. Out of 25 students, find the probability that exactly 18 pass.

6 answer → Calculator **Y13 Worksheet · Q4**

[Maximum mark: 2]

A machine produces 5% defective items. Out of a sample of 40, find the probability of getting fewer than 3 defective items.

7 answer → Calculator **Oxford Ex 14D · Q1**

[Maximum mark: 3]

A regular tetrahedron has three white faces and one red face. It is rolled four times and the colour of the bottom face is noted. Find the most likely number of times that the red face will end downwards. Calculate the probability of this value occurring.

8 answer → Calculator **Oxford Ex 14D · Q2**

[Maximum mark: 4]

The probability that a marksman scores a bull's eye when he shoots at a target is 0.55. Find the probability that in eight attempts:

(a) He hits the bull's eye five times. [2]

(b) He misses the bull's eye at least five times. [2]

5 ← question *Y13 Worksheet · Q3*

$X \sim \mathrm{B}(25, 0.8)$; $\mathrm{P}(X = 18) = \boxed{0.111}$ (0.110842) **M1 A1**

(The answer key printed 0.148 — this is not correct; the GDC gives 0.110842.)

TI-Nspire CX: Question 5

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(25,0.8,18)`

6 ← question *Y13 Worksheet · Q4*

$X \sim \mathrm{B}(40, 0.05)$; $\mathrm{P}(X < 3) = \mathrm{P}(X \leq 2) = \boxed{0.677}$ **M1 A1**

TI-Nspire CX: Question 6

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(40,0.05,0,2)`

7 ← question *Oxford Ex 14D · Q1*

$X \sim \mathrm{B}(4, 0.25)$: $\mathrm{P}(X = 0) = 0.316$, $\mathrm{P}(X = 1) = 0.422$, $\mathrm{P}(X = 2) = 0.211, \ldots$ **M1**

Most likely: $\boxed{X = 1}$ with probability $\boxed{0.422}$ **A1 A1**

TI-Nspire CX: Question 7

Lists & Spreadsheet: column A $= 0, 1, 2, 3, 4$; in B type `=binompdf(4,0.25,a1)` and fill down, or use **Binomial Pdf** with the X value left blank to get the whole list.

8 ← question *Oxford Ex 14D · Q2*

$X \sim \mathrm{B}(8, 0.55)$ = hits

(a) $\mathrm{P}(X = 5) = \boxed{0.257}$ **A2**

(b) misses $\geq 5 \Leftrightarrow$ hits ≤ 3: $\mathrm{P}(X \leq 3) = \boxed{0.260}$ **M1 A1**

TI-Nspire CX: Question 8

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(8,0.55,5)`; `binomCdf(8,0.55,0,3)`

9 answer → Calculator **Oxford Ex 14D · Q3**

[Maximum mark: 5]

A factory has four machines making the same type of component. The probability that any machine will produce a faulty component is 0.01. Find the probability that, in a sample of four components from each machine:

(a) None will be faulty. [1]

(b) Exactly 13 will not be faulty. [2]

(c) At least two will be faulty. [2]

10 answer → Calculator **Oxford Ex 14D · Q4**

[Maximum mark: 4]

The probability that a single telephone line is engaged at a company switchboard is 0.25. If the switchboard has 10 lines, find the probability that:

(a) One half of the lines are engaged. [2]

(b) At least three lines are free. [2]

11 answer → Calculator **Oxford Ex 14D · Q5**

[Maximum mark: 2]

The probability that Nicole goes to bed at 7.30 pm on a given day is 0.4. Calculate the probability that, on five consecutive days, Nicole goes to bed at 7.30 pm on at most three days.

9 ← question *Oxford Ex 14D · Q3*

16 components: $X \sim \mathrm{B}(16, 0.01)$ = number faulty.

(a) $\mathrm{P}(X = 0) = 0.99^{16} = \boxed{0.851}$ **A1**

(b) 13 not faulty $\Rightarrow$ 3 faulty: $\mathrm{P}(X = 3) = \boxed{0.000491}$ **M1 A1**

(c) $\mathrm{P}(X \geq 2) = \boxed{0.0109}$ **M1 A1**

TI-Nspire CX: Question 9

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).
`binomPdf(16,0.01,3); binomCdf(16,0.01,2,16)`

10 ← question *Oxford Ex 14D · Q4*

$X \sim \mathrm{B}(10, 0.25)$ = engaged lines

(a) $\mathrm{P}(X = 5) = \boxed{0.0584}$ **A2**

(b) free $= 10 - X \geq 3 \Leftrightarrow X \leq 7$: $\mathrm{P}(X \leq 7) = \boxed{0.9996}$ (≈ 1.00) **M1 A1**

TI-Nspire CX: Question 10

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).
`binomPdf(10,0.25,5); binomCdf(10,0.25,0,7)`

11 ← question *Oxford Ex 14D · Q5*

$X \sim \mathrm{B}(5, 0.4)$; $\mathrm{P}(X \leq 3) = \boxed{0.913}$ **M1 A1**

TI-Nspire CX: Question 11

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(5,0.4,0,3)`

12 answer → Calculator **Oxford Ex 14D · Q6**

[Maximum mark: 4]

In an examination hall, it is known that 15% of desks are wobbly.

(a) What is the probability that, in a row of six desks, more than one will be wobbly? [2]

(b) What is the probability that exactly one desk will be wobbly in a row of six desks? [2]

12 ← question *Oxford Ex 14D · Q6*

$X \sim \mathrm{B}(6, 0.15)$

(a) $\mathrm{P}(X > 1) = \mathrm{P}(X \geq 2) = \boxed{0.224}$ **M1 A1**

(b) $\mathrm{P}(X = 1) = \boxed{0.399}$ **A2**

TI-Nspire CX: Question 12

`binomCdf(6,0.15,2,6);` menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: *n*, *p*, X value (one value). `binomPdf(6,0.15,1)`

13 answer → Calculator **Oxford Ex 14D · Q7**

[Maximum mark: 9]

In the mass production of computer processors, it is found that 5% are defective. Processors are selected at random and put into packs of 15.

(a) If a pack is selected at random, find the probability that it will contain: (i) three defective processors (ii) no defective processors (iii) at least two defective processors. [4]

(b) Two packs are selected at random. Find the probability that there are: (i) no defective processors in either pack (ii) at least two defective processors in either pack (iii) no defective processors in one pack and at least two in the other. [5]

13 ← question *Oxford Ex 14D · Q7*

$X \sim \mathrm{B}(15, 0.05)$

(a)(i) $\mathrm{P}(X = 3) = \boxed{0.0307}$ (ii) $\mathrm{P}(X = 0) = \boxed{0.463}$ (iii) $\mathrm{P}(X \geq 2) = \boxed{0.171}$ **A1 A1 A2**

(b)(i) $0.463291^2 = \boxed{0.215}$ (ii) $0.170953^2 = \boxed{0.0292}$ **A1 A1**

(iii) either pack can be the one with none: $2 \times 0.463291 \times 0.170953 = \boxed{0.158}$ **M1 M1 A1**

TI-Nspire CX: Question 13

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(15,0.05,3)`; `binomCdf(15,0.05,2,15)`. Store each result (ctrl var) before squaring.

14 answer →

Calculator SME Medium · Q2

[Maximum mark: 3]

(a) A fair coin is tossed 20 times and the number of times it lands heads up is recorded. Find the expected number of times that the coin will land heads up. [1]

(b) Find the probability that the coin lands heads up 15 times. [2]

15 answer →

Calculator SME Medium · Q4

[Maximum mark: 4]

(a) A difficult operation to remove a rare form of cancer is found to have a success rate of 78%. Twelve patients are on the waiting list to undergo the operation. Find the probability that all twelve patients' operations are successful. [2]

(b) Find the probability that all but two patients undergo successful operations. [2]

16 answer →

Calculator SME Medium · Q5

[Maximum mark: 4]

For a jellyfish population in a certain area of the ocean, there is a 95% chance that any given jellyfish contains microplastic particles in its body. For a sample size of 40 jellyfish from this population, find the probability of: (i) exactly 38 jellyfish (ii) all the jellyfish having microplastic particles in their bodies.

14 ← question *SME Medium · Q2*

$X \sim \mathrm{B}(20, 0.5)$

(a) $\mathrm{E}(X) = 20 \times 0.5 = \boxed{10}$ **A1**

(b) $\mathrm{P}(X = 15) = \boxed{0.0148}$ **M1 A1**

TI-Nspire CX: Question 14

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).

```
binomPdf(20,0.5,15)
```

15 ← question *SME Medium · Q4*

$X \sim \mathrm{B}(12, 0.78)$

(a) $\mathrm{P}(X = 12) = 0.78^{12} = \boxed{0.0507}$ **M1 A1**

(b) $\mathrm{P}(X = 10) = \boxed{0.266}$ **M1 A1**

TI-Nspire CX: Question 15

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).

```
binomPdf(12,0.78,10)
```

16 ← question *SME Medium · Q5*

$X \sim \mathrm{B}(40, 0.95)$

(i) $\mathrm{P}(X = 38) = \boxed{0.278}$ **M1 A1**

(ii) $\mathrm{P}(X = 40) = 0.95^{40} = \boxed{0.129}$ **M1 A1**

TI-Nspire CX: Question 16

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).

```
binomPdf(40,0.95,38)
```

17 answer →

Calculator SME Medium · Q7

[Maximum mark: 4]

For cans of a particular brand of soft drink labelled as containing 330 ml, the actual volume of soft drink in a can varies. Although the company's quality control assures that the mean volume remains at 330 ml, it is known from experience that the probability of any particular can containing less than 320 ml is 0.0296.

Tilly buys a pack of 24 cans of this soft drink. It may be assumed that those cans represent a random sample. Let L represent the number of cans in the pack that contain less than 320 ml of soft drink. Find the probability that (i) none of the cans (ii) exactly two of the cans (iii) at least two of the cans contain less than 320 ml of soft drink.

18 answer →

Calculator SME Medium · Q8

[Maximum mark: 7]

The random variable $X \sim \mathrm{B}(40, 0.15)$.

(a) Find (i) $\mathrm{P}(3 \leq X < 14)$ (ii) $\mathrm{P}(5 < X < 12)$ [3]

(b) Find $\mathrm{Var}(X)$. [1]

(c) Find $\mathrm{P}(X \leq 3 \,|\, X \leq 9)$. [3]

17 ← question *SME Medium · Q7*

$L \sim \mathrm{B}(24, 0.0296)$

(i) $\mathrm{P}(L = 0) = \boxed{0.486}$ (ii) $\mathrm{P}(L = 2) = \boxed{0.125}$ (iii) $\mathrm{P}(L \geq 2) = \boxed{0.158}$ **A1 A1 A2**

TI-Nspire CX: Question 17

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).

`binomPdf(24,0.0296,2); binomCdf(24,0.0296,2,24)`

18 ← question *SME Medium · Q8*

(a)(i) $\mathrm{P}(3 \leq X \leq 13) = \boxed{0.950}$ (ii) $\mathrm{P}(6 \leq X \leq 11) = \boxed{0.556}$ **M1 A1 A1**

(b) $40(0.15)(0.85) = \boxed{5.1}$ **A1**

(c) $\dfrac{\mathrm{P}(X \leq 3)}{\mathrm{P}(X \leq 9)} = \dfrac{0.130168}{0.932784} = \boxed{0.140}$ **M1 M1 A1**

Discrete: < 14 means ≤ 13; > 5 means ≥ 6.

TI-Nspire CX: Question 18

`binomCdf(40,0.15,3,13), binomCdf(40,0.15,6,11), binomCdf(40,0.15,0,3)/binomCdf(40,0.15,0,`

19 answer →

Calculator SME Hard · Q4

[Maximum mark: 7]

In a normal pack of 52 playing cards there are 13 of each suit: hearts, clubs, diamonds and spades. James plays a card game with some friends. Each player is given nine full packs of cards. They randomly pick one card from each full pack so that they each have a hand of nine cards.

(a) Find the probability that James has at least three clubs. [2]

(b) Hearts and diamonds are the red suits; clubs and spades are the black suits. Find the probability that James only has red cards in his hand. [3]

(c) Find the probability that James has twice as many black cards as red cards in his hand. [2]

20 answer →

Calculator Kognity B · Q5

[Maximum mark: 3]

In a multiple choice test, there are 10 questions. Each question has 4 choices, one of which is correct. If 60% is the pass mark and Tom, who knows nothing about the subject, guesses each answer at random, determine the probability that he will **not** pass.

1. 0.98 **2.** 0.68 **3.** 0.78 **4.** 0.88

21 answer →

No calculator SME Hard · Q11

[Maximum mark: 6]

(a) A fair coin is tossed seven times. Find the probability of getting exactly five tails. Give your answer as a fraction. [3]

(b) Find the probability of getting exactly three heads. Give your answer as a fraction. [3]

19 ← question *SME Hard · Q4*

(a) $C \sim \mathrm{B}(9, 0.25)$: $\mathrm{P}(C \geq 3) = \boxed{0.399}$ **M1 A1**

(b) $R \sim \mathrm{B}(9, 0.5)$: $\mathrm{P}(R = 9) = 0.5^9 = \boxed{0.00195}$ $\left(\dfrac{1}{512}\right)$ **M1 A2**

(c) 6 black, 3 red: $\mathrm{P}(R = 3) = \boxed{0.164}$ **M1 A1**

TI-Nspire CX: Question 19

`binomCdf(9,0.25,3,9);` menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(9,0.5,3)`

20 ← question *Kognity B · Q5*

$X \sim \mathrm{B}(10, 0.25)$; pass needs $X \geq 6$. Not pass: $\mathrm{P}(X \leq 5) = 0.9803$ **M1 A1**

Option 1: 0.98 **A1**

TI-Nspire CX: Question 20

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(10,0.25,0,5)`

21 ← question *SME Hard · Q11*

(a) $\binom{7}{5}\left(\dfrac{1}{2}\right)^7 = \dfrac{21}{128}$ **M1 A1** $\boxed{\dfrac{21}{128}}$ **A1**

(b) $\binom{7}{3}\left(\dfrac{1}{2}\right)^7 = \boxed{\dfrac{35}{128}}$ **M1 A1 A1**

22 answer →

No calculator

SME Hard · Q12

[Maximum mark: 6]

(a) On any given day, the probability that Mark rides his bike to school is $\frac{1}{3}$. Find the probability that, in a week with five school days, Mark rides his bike to school on exactly two days. Give your answer as a fraction. [3]

(b) Find the probability that, in a week with five school days, Mark rides his bike to school on at least one day. [3]

22 ← question *SME Hard · Q12*

$X \sim \mathrm{B}\left(5, \frac{1}{3}\right)$

(a) $\binom{5}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^3 = 10 \cdot \frac{1}{9} \cdot \frac{8}{27} = \boxed{\frac{80}{243}}$ **M1 A1 A1**

(b) $1 - \left(\frac{2}{3}\right)^5 = 1 - \frac{32}{243} = \boxed{\frac{211}{243}}$ **M1 A1 A1**

23 answer → Calculator **Kognity B · Q1**

[Maximum mark: 1]

The random variable X follows a binomial distribution with parameters $n = 138$ and $p = 0.17$. Find the expected value of this random variable. Give an exact answer, do not round.

24 answer → Calculator **Kognity C · Q4**

[Maximum mark: 2]

Marina has learned that it typically rains 43% of the days in August in the city where she lives. Assuming that the probability of rain on each day is independent, find the expected number of days it would rain in Marina's city in two weeks during the month of August. *Round your answer to the nearest whole number of days.*

25 answer → Calculator **Kognity D · Q3**

[Maximum mark: 2]

Marina has learned that it typically rains 43% of the days in August in the city where she lives. Assuming that the probability of rain on each day is independent, find the variance in the number of rainy days in Marina's city in two weeks during the month of August. Round your answer to three significant figures.

26 answer → Calculator **Y13 Worksheet · Q5**

[Maximum mark: 3]

The probability that a component fails in its first month is 0.04. In a batch of 80, find the expected number of failures and the standard deviation.

23 ← question *Kognity B · Q1*

$$\mathrm{E}(X) = np = 138 \times 0.17 = \boxed{23.46}$$ **A1**

(Do not round to 23: an expected value need not be a whole number.)

24 ← question *Kognity C · Q4*

$X \sim \mathrm{B}(14, 0.43)$; $\mathrm{E}(X) = 14 \times 0.43 = 6.02$ **M1** $\approx \boxed{6}$ days **A1**

(Here the question asks for a whole number; otherwise leave $\mathrm{E}(X) = 6.02$ unrounded.)

25 ← question *Kognity D · Q3*

$\mathrm{Var}(X) = np(1-p) = 14(0.43)(0.57) = \boxed{3.43}$ **M1 A1**

26 ← question *Y13 Worksheet · Q5*

$X \sim \mathrm{B}(80, 0.04)$: $\mathrm{E}(X) = 80 \times 0.04 = \boxed{3.2}$ **A1**

$\sigma = \sqrt{80(0.04)(0.96)} = \sqrt{3.072} = \boxed{1.75}$ **M1 A1**

27 answer →

Calculator SME Medium · Q3

[Maximum mark: 5]

(a) On any given day during a normal five day working week, there is a 60% chance that Yussuf catches a taxi to work. Find $\mathrm{E}(X)$, the expected number of times Yussuf will catch a taxi to work during a normal five day working week. [1]

(b) Find the probability that, during a normal five-day working week, Yussuf never catches a taxi. [2]

(c) Find the probability that, during a normal five-day working week, Yussuf catches a taxi once at the most. [2]

28 answer →

Calculator RV · Q6

[Maximum mark: 6]

A multiple choice exam consists of 20 questions. Each question has four answers. Only one of the answers is correct. For each question, Jack randomly chooses one of the four answers.

(a) Find the expected number of questions Jack answers correctly. [2]

(b) Find the probability that Jack: (i) answers exactly four questions correctly; (ii) answers more than four questions correctly. [4]

27 ← question *SME Medium · Q3*

$X \sim \mathrm{B}(5, 0.6)$

(a) $\boxed{3}$ **A1**

(b) $0.4^5 = \boxed{0.01024}$ **M1 A1**

(c) $\mathrm{P}(X \leq 1) = \boxed{0.0870}$ **M1 A1**

TI-Nspire CX: Question 27

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(5,0.6,0,1)`

28 ← question *RV · Q6*

$X \sim \mathrm{B}\left(20, \frac{1}{4}\right)$

(a) $\mathrm{E}(X) = 20 \times \frac{1}{4} = \boxed{5}$ **M1 A1**

(b)(i) $\mathrm{P}(X = 4) = \boxed{0.190}$ **M1 A1**

(ii) $\mathrm{P}(X > 4) = \mathrm{P}(X \geq 5) = \boxed{0.585}$ **M1 A1**

TI-Nspire CX: Question 28

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(20,0.25,4); binomCdf(20,0.25,5,20)`

29 answer →

Calculator **Oxford Ex 14G · Q2**

[Maximum mark: 5]

A multiple choice test has 20 questions. There are five possible answers to each question, where only one is correct. If X is "the number of questions a student guesses correctly", find:

(a) the distribution of X [1]

(b) the mean and variance of X [2]

(c) the probability that, if a student answers the test purely by guessing, they will achieve the pass mark of 10 or more. [2]

30 answer →

Calculator **RV · Q28**

[Maximum mark: 6]

The probability of obtaining tails when a biased coin is tossed is 0.6. The coin is tossed 400 times.

(a) (i) Write down the mean number of tails. (ii) Find the variance of the number of tails. [3]

(b) Find the probability that the number of tails obtained is less than one standard deviation away from the mean. [3]

29 ← question *Oxford Ex 14G · Q2*

(a) $\boxed{X \sim \mathrm{B}(20, 0.2)}$ **A1**

(b) $\mathrm{E}(X) = \boxed{4}$, $\mathrm{Var}(X) = 20(0.2)(0.8) = \boxed{3.2}$ **A1 A1**

(c) $\mathrm{P}(X \geq 10) = \boxed{0.00259}$ **M1 A1**

TI-Nspire CX: Question 29

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(20,0.2,10,20)`

30 ← question *RV · Q28*

$T \sim \mathrm{B}(400, 0.6)$

(a)(i) $\mathrm{E}(T) = \boxed{240}$ (ii) $\mathrm{Var}(T) = 400(0.6)(0.4) = \boxed{96}$ **A1 M1 A1**

(b) $\sigma = \sqrt{96} = 9.80$; $\mathrm{P}(230.2 < T < 249.8) = \mathrm{P}(231 \leq T \leq 249)$ **M1 M1**

$= \boxed{0.668}$ **A1**

Discrete: round the end values **inwards** to whole numbers.

TI-Nspire CX: Question 30

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(400,0.6,231,249)`

31 answer → Calculator **SME Medium · Q9**

[Maximum mark: 15]

(a) Zara is a gymnast. It is known that she has a 20% chance of making a mistake in any given routine. Zara performs ten routines in a competition. (i) Find the expected number of routines in which Zara will make a mistake. (ii) Find the standard deviation of the number of routines in which Zara makes a mistake. [3]

(b) Find the probability that Zara makes a mistake in: (i) none of her routines (ii) two of her routines (iii) no more than two of her routines. [6]

(c) Find the probability that Zara makes a mistake in 3 of her routines, given that she makes a mistake in at least 2 of her routines. [3]

(d) Find the probability that the number of routines in which Zara makes a mistake is less than one standard deviation away from the mean. [3]

31 ← question *SME Medium · Q9*

$X \sim \mathrm{B}(10, 0.2)$

(a)(i) $\boxed{2}$ (ii) $\sqrt{10(0.2)(0.8)} = \sqrt{1.6} = \boxed{1.26}$ **A1 M1 A1**

(b)(i) $0.8^{10} = \boxed{0.107}$ (ii) $\boxed{0.302}$ (iii) $\mathrm{P}(X \leq 2) = \boxed{0.678}$ **A2 A2 A2**

(c) $\dfrac{\mathrm{P}(X=3)}{\mathrm{P}(X \geq 2)} = \dfrac{0.201327}{0.624190} = \boxed{0.323}$ **M1 M1 A1**

(d) $2 - 1.26 < X < 2 + 1.26 \Rightarrow 0.735 < X < 3.26 \Rightarrow 1 \leq X \leq 3$: $\boxed{0.772}$

M1 M1 A1

TI-Nspire CX: Question 31

`binomPdf(10,0.2,2)`, `binomCdf(10,0.2,0,2)`, `binomCdf(10,0.2,1,3)`

32 answer →

Calculator **SME Medium · Q10**

[Maximum mark: 8]

(a) In the town of Wooster, Ohio, it is known that 90% of the residents prefer the locally produced Woostershire brand sauce when preparing a Caesar salad. The other 10% prefer another well-known brand. 30 residents are chosen at random by a pollster. Let X represent the number of those 30 residents that prefer Woostershire brand sauce. Find (i) $\mathrm{E}(X)$ (ii) $\mathrm{Var}(X)$ [2]

(b) Find the probability that (i) 90% or more of the residents chosen prefer Woostershire brand sauce (ii) none of the residents chosen prefer the other well-known brand. [6]

33 answer →

Calculator **SME Hard · Q1**

[Maximum mark: 7]

(a) The germination rate of a particular seed is 44%. George sows 25 of these seeds selected at random. Calculate the expected number of seeds that germinate. [2]

(b) Find the probability that more than 7 of these seeds will germinate. [2]

(c) Find the probability that at least 9, but no more than 11 of the seeds germinate. [3]

32 ← question *SME Medium · Q10*

$X \sim \mathrm{B}(30, 0.9)$

(a)(i) $\boxed{27}$ (ii) $30(0.9)(0.1) = \boxed{2.7}$ **A1 A1**

(b)(i) 90% of 30 = 27: $\mathrm{P}(X \geq 27) = \boxed{0.647}$ **M1 A2**

(ii) all 30 prefer Woostershire: $0.9^{30} = \boxed{0.0424}$ **M1 A2**

TI-Nspire CX: Question 32

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(30,0.9,27,30)`

33 ← question *SME Hard · Q1*

$X \sim \mathrm{B}(25, 0.44)$

(a) $25 \times 0.44 = \boxed{11}$ **M1 A1**

(b) $\mathrm{P}(X \geq 8) = \boxed{0.923}$ **M1 A1**

(c) $\mathrm{P}(9 \leq X \leq 11) = \boxed{0.426}$ **M1 A2**

TI-Nspire CX: Question 33

`binomCdf(25,0.44,8,25); binomCdf(25,0.44,9,11)`

34 answer →

Calculator **SME Hard · Q2**

[Maximum mark: 6]

(a) At a builders' convention there is a competition in which 17 contestants get to choose one out of ten gift boxes. Two of the gift boxes have a chainsaw and the others have a regular saw. After a builder selects a box and claims the gift, the saw is replenished so that the next builder has the same options. Find the probability that a contestant gets a regular saw. [1]

(b) Find the probability that exactly five of the 17 contestants win a chainsaw. [2]

(c) Find the probability that at least three and no more than nine contestants win a chainsaw. [3]

35 answer →

Calculator **SME Hard · Q3**

[Maximum mark: 8]

(a) A biased dice is rolled 24 times. The probability that the dice lands on a prime number is $\frac{3}{4}$. The dice is equally likely to land on any of the prime numbers. Similarly, the other numbers all have the same chance as each other of being the number that the dice lands on. Find the probability that the dice lands on a prime number no more than 15 times. [2]

(b) Find the expected number of times the dice lands on a three. [3]

(c) Find the expected number of times the dice lands on a one. [3]

34 ← question *SME Hard · Q2*

(a) $\frac{8}{10} = \boxed{0.8}$ **A1**

$X \sim \mathrm{B}(17, 0.2)$ = chainsaws

(b) $\mathrm{P}(X = 5) = \boxed{0.136}$ **M1 A1**

(c) $\mathrm{P}(3 \le X \le 9) = \boxed{0.690}$ **M1 A2**

TI-Nspire CX: Question 34

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(17,0.2,5); binomCdf(17,0.2,3,9)`

35 ← question *SME Hard · Q3*

(a) $X \sim \mathrm{B}(24, 0.75)$: $\mathrm{P}(X \le 15) = \boxed{0.121}$ **M1 A1**

(b) primes 2, 3, 5 share $\frac{3}{4}$: $\mathrm{P}(3) = \frac{1}{4}$; $24 \times \frac{1}{4} = \boxed{6}$ **M1 M1 A1**

(c) 1, 4, 6 share $\frac{1}{4}$: $\mathrm{P}(1) = \frac{1}{12}$; $24 \times \frac{1}{12} = \boxed{2}$ **M1 M1 A1**

TI-Nspire CX: Question 35

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(24,0.75,0,15)`

36 answer → Calculator **SME Hard · Q5**

[Maximum mark: 6]

(a) John has a collection of 12 suit jackets, of which 5 are tailored to fit him perfectly. On any given day, there is a 90% chance that John will wear a suit jacket and if he does, he chooses the suit jacket randomly. Find the probability that on a given day John wears a tailored suit jacket. [2]

(b) In a year John works 260 days. Find the expected number of days in a year that John wears a suit jacket. [1]

(c) Find the probability that John wears a non-tailored suit jacket to work at least 150 days in a year. [3]

36 ← question *SME Hard · Q5*

(a) $0.9 \times \dfrac{5}{12} = \boxed{0.375}$ **M1 A1**

(b) $260 \times 0.9 = \boxed{234}$ days **A1**

(c) $p = 0.9 \times \dfrac{7}{12} = 0.525$; $X \sim \mathrm{B}(260, 0.525)$: $\mathrm{P}(X \geq 150) = \boxed{0.0529}$

M1 M1 A1

TI-Nspire CX: Question 36

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(260,0.525,150,260)`

37 answer → Calculator SME Hard · Q6

[Maximum mark: 6]

(a) Two fair dice are rolled and the numbers showing on the dice are added together. This is done 15 times and the number of times the sum is equal to 2, 7 or 11 is recorded. Let X be the number of times that the sum is equal to 2, 7 or 11. Find the expected value of X. [3]

(b) Find the probability that X is at least 2 but no more than 6. [3]

38 answer → Calculator SME Hard · Q7

[Maximum mark: 9]

(a) In a game show, each contestant has seven boxes they can open. Two boxes are empty, two boxes have \$10 inside, two boxes have \$100 inside and one box has \$10 000 inside. A contestant randomly selects a box and wins the amount that is inside. Find the expected payoff from playing the game. [2]

(b) Suppose there are 28 contestants. Find the expected number of players that win (i) \$10 (ii) \$100 (iii) \$10 000. [3]

(c) Find the probability that (i) exactly four contestants win \$10 000 (ii) less than six contestants win \$10 000. [4]

37 ← question *SME Hard · Q6*

(a) sums 2, 7, 11: $1+6+2=9$ of 36 outcomes, $p=\dfrac{1}{4}$ **M1 A1**

$\mathrm{E}(X)=15\times\dfrac{1}{4}=\boxed{3.75}$ **A1**

(b) $X\sim\mathrm{B}(15,0.25)$: $\mathrm{P}(2\le X\le 6)=\boxed{0.863}$ **M1 A2**

TI-Nspire CX: Question 37

menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: n, p, Lower, Upper (a range, both ends included). `binomCdf(15,0.25,2,6)`

38 ← question *SME Hard · Q7*

(a) $\dfrac{1}{7}(0\cdot 2+10\cdot 2+100\cdot 2+10000)=\dfrac{10220}{7}=\$\boxed{1460}$ **M1 A1**

(b)(i) $28\times\dfrac{2}{7}=\boxed{8}$ (ii) $\boxed{8}$ (iii) $28\times\dfrac{1}{7}=\boxed{4}$ **A1 A1 A1**

(c) $X\sim\mathrm{B}\left(28,\dfrac{1}{7}\right)$: (i) $\mathrm{P}(X=4)=\boxed{0.211}$ (ii) $\mathrm{P}(X\le 5)=\boxed{0.798}$ **M1 A1 M1 A1**

TI-Nspire CX: Question 38

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(28,1/7,4); binomCdf(28,1/7,0,5)`

39 answer → Calculator **SME Hard · Q8**

[Maximum mark: 8]

(a) Greg sells A4 paintings for \$14.75 each at a market. The probability that someone at the market buys a painting from his stall is 0.06. On a given day there are 250 people at the market. Find the expected number of people that will buy a painting from Greg's stall. [2]

(b) Find the probability that 20 or more people buy a painting from Greg's stall. [2]

(c) Find the probability that Greg earns more than \$118. [4]

39 ← question *SME Hard · Q8*

$X \sim \mathrm{B}(250, 0.06)$

(a) $\boxed{15}$ **M1 A1**

(b) $\mathrm{P}(X \geq 20) = \boxed{0.118}$ **M1 A1**

(c) $14.75X > 118 \Rightarrow X > 8 \Rightarrow X \geq 9$ **M1 A1**

$\mathrm{P}(X \geq 9) = \boxed{0.967}$ **M1 A1**

TI-Nspire CX: Question 39

```
binomCdf(250,0.06,20,250); binomCdf(250,0.06,9,250)
```

40 answer → Calculator **SME Hard · Q9**

[Maximum mark: 6]

(a) Scott works as a real estate agent. The probability that he sells a house to someone over the age of 30 is 0.11 and the probability he sells a house to someone 30 or under is 0.04. In a given week, Scott interacts with 32 potential buyers, 17 of which are over 30 and the rest are 30 or under. Find the expected number of houses that Scott sells in a week. [3]

(b) Find the probability Scott sells at least one house in a week. [3]

41 answer → Calculator **SME Hard · Q10**

[Maximum mark: 8]

(a) A multiple-choice test consists of 10 questions, where only one option is correct; six questions have three possible options and the other four questions have four possible options. Isaac takes the test and randomly selects an answer for each question. Find the expected number of questions Isaac answers correctly. [3]

(b) Find the probability that Isaac gets exactly two answers correct. [5]

40 ← question *SME Hard · Q9*

(a) $17(0.11)+15(0.04)=1.87+0.6=\boxed{2.47}$ **M1 M1 A1**

(b) $1-\text{P}(\text{none})=1-0.89^{17}\times 0.96^{15}=\boxed{0.925}$ **M1 M1 A1**

(Two different binomials: $\text{B}(17,0.11)$ and $\text{B}(15,0.04)$.)

41 ← question *SME Hard · Q10*

(a) $6\times\frac{1}{3}+4\times\frac{1}{4}=\boxed{3}$ **M1 M1 A1**

(b) $A\sim\text{B}(6,\frac{1}{3})$, $B\sim\text{B}(4,\frac{1}{4})$: $\text{P}(A{=}2)\text{P}(B{=}0)+\text{P}(A{=}1)\text{P}(B{=}1)+\text{P}(A{=}0)\text{P}(B{=}2)$ **M1 M1**

$=0.329218(0.316406)+0.263374(0.421875)+0.087791(0.210938)=\boxed{0.234}$ **A1 A1 A1**

TI-Nspire CX: Question 41

`binomPdf(6,1/3,k)` and `binomPdf(4,1/4,k)` for $k=0,1,2$.

42 answer → Calculator **Y13 Worksheet · Q6**

[Maximum mark: 4]

In a game, the probability of scoring a goal is p. In 12 attempts, the probability of scoring at least 2 goals is 0.99. Find the smallest possible value of p.

43 answer → Calculator **Oxford Ex 14E · Q1**

[Maximum mark: 3]

If $X \sim \mathrm{B}(n, 0.6)$ and $\mathrm{P}(X < 1) = 0.0256$, find n.

44 answer → Calculator **Oxford Ex 14E · Q2**

[Maximum mark: 3]

1% of fuses in a large box of fuses are faulty. What is the largest sample size that can be taken if it is required that the probability that there are no faulty fuses in the sample is greater than 0.5?

45 answer → Calculator **Oxford Ex 14E · Q3**

[Maximum mark: 3]

If $X \sim \mathrm{B}(n, 0.2)$ and $\mathrm{P}(X \geq 1) > 0.75$, find the least possible value of n.

42 ← question *Y13 Worksheet · Q6*

$X \sim \mathrm{B}(12, p)$: $1 - \mathrm{P}(X = 0) - \mathrm{P}(X = 1) = 0.99$ **M1**

$(1-p)^{12} + 12p(1-p)^{11} = 0.01$ **A1**

$\boxed{p = 0.440}$ (by GDC) **A2**

(The printed answer key gives 0.21 — check: $p = 0.21$ gives only 0.74.)

TI-Nspire CX: Question 42

Graphs: `f1(x)=1-binomCdf(12,x,0,1)`, `f2(x)=0.99`; window $0 \le x \le 1$; menu → 6: Analyze → 4: Intersection → $x = 0.440$.

43 ← question *Oxford Ex 14E · Q1*

$\mathrm{P}(X = 0) = 0.4^n = 0.0256$ **M1**

$n = \dfrac{\ln 0.0256}{\ln 0.4} = \boxed{4}$ **A1 A1**

44 ← question *Oxford Ex 14E · Q2*

$0.99^n > 0.5 \Rightarrow n < \dfrac{\ln 0.5}{\ln 0.99} = 68.97$ **M1 A1**

largest $\boxed{n = 68}$ **A1**

TI-Nspire CX: Question 44

Table: ctrl T on a Graphs page with `f1(x)=...`, scroll to the first x that works. `f1(x)=0.99^x`.

45 ← question *Oxford Ex 14E · Q3*

$1 - 0.8^n > 0.75 \Rightarrow 0.8^n < 0.25 \Rightarrow n > 6.21$ **M1 A1**

$\boxed{n = 7}$ **A1**

TI-Nspire CX: Question 45

Table: ctrl T on a Graphs page with `f1(x)=...`, scroll to the first x that works. `f1(x)=1-0.8^x`.

46 answer → Calculator **Oxford Ex 14E · Q4**

[Maximum mark: 3]

The probability that Anna scores a penalty goal in a hockey competition is 0.3. Find the least number of penalty attempts that she would need to make if the probability that she scores a goal at least once is greater than 0.95.

47 answer → Calculator **Oxford Ex 14E · Q5**

[Maximum mark: 3]

Find the least number of times an unbiased coin must be tossed so that the probability that at least one tail will occur is greater than or equal to 0.99.

48 answer → Calculator **Oxford Ex 14G · Q1**

[Maximum mark: 4]

X is a discrete random variable where $X \sim \mathrm{B}(n, p)$, $\mathrm{E}(X) = 12$ and $\mathrm{Var}(X) = 3$. Find n and p.

49 answer → No calculator **Oxford Exam-style · Q20**

[Maximum mark: 5]

A random variable X follows a binomial distribution with mean 3 and variance 1.2. Find the value of the parameters n and p of the distribution of X.

46 ← question *Oxford Ex 14E · Q4*

$1-0.7^n > 0.95 \Rightarrow 0.7^n < 0.05 \Rightarrow n > 8.40$ **M1 A1**

$\boxed{n = 9}$ **A1**

TI-Nspire CX: Question 46

Table: ctrl T on a Graphs page with `f1(x)=...`, scroll to the first x that works. `f1(x)=1-0.7^x`.

47 ← question *Oxford Ex 14E · Q5*

$1-0.5^n \geq 0.99 \Rightarrow 0.5^n \leq 0.01 \Rightarrow n \geq 6.64$ **M1 A1**

$\boxed{n = 7}$ **A1**

TI-Nspire CX: Question 47

Table: ctrl T on a Graphs page with `f1(x)=...`, scroll to the first x that works. `f1(x)=1-0.5^x`.

48 ← question *Oxford Ex 14G · Q1*

$np = 12,\ np(1-p) = 3 \Rightarrow 1-p = \dfrac{3}{12} = \dfrac{1}{4}$ **M1 M1**

$\boxed{p = \dfrac{3}{4}}$, $\boxed{n = 16}$ **A1 A1**

49 ← question *Oxford Exam-style · Q20*

$np = 3,\ npq = 1.2$ **M1 A1**

$q = \dfrac{1.2}{3} = 0.4 \Rightarrow \boxed{p = 0.6}$ **A1**

$n = \dfrac{3}{0.6} = \boxed{5}$ **A1 A1**

50 answer →

Calculator **Oxford Ex 14G · Q3**

[Maximum mark: 4]

A random variable X has the distribution $\mathrm{B}(12, p)$. Given that the variance of X is 1.92, find the possible values of p.

51 answer →

Calculator **RV · Q43**

[Maximum mark: 6]

A bag contains 8 blue marbles, 12 green marbles and m red marbles. A marble is selected at random and replaced. This is performed three times.

(a) Write down the probability that the first marble selected is blue. [1]

(b) Let X be the number of blue marbles selected. Find the smallest value of m for which $\mathrm{Var}(X) < 0.5$. [5]

52 answer →

No calculator **SME Very Hard · Q11**

[Maximum mark: 8]

(a) A biased dice is thrown 44 times and the expected number of times it will land on a six is 33. The dice is thrown another eight times. Find the expected number of sixes seen from the eight throws. [2]

(b) Calculate the variance of the number of sixes seen from the eight throws. Give your answer as a fraction. [2]

(c) Find the probability that exactly four sixes are seen from the eight throws. Give your answer as a fraction. [4]

50 ← question *Oxford Ex 14G · Q3*

$12p(1-p) = 1.92 \Rightarrow p^2 - p + 0.16 = 0$ **M1 A1**

$(p-0.8)(p-0.2) = 0$: $\boxed{p = 0.2 \text{ or } 0.8}$ **A1 A1**

TI-Nspire CX: Question 50

menu → **3: Algebra** → **1: Solve**. `solve(12p(1-p)=1.92,p)`.

51 ← question *RV · Q43*

(a) $\text{P(blue)} = \boxed{\dfrac{8}{20+m}}$ **A1**

(b) $X \sim \text{B}\left(3, \dfrac{8}{20+m}\right)$: $\text{Var}(X) = 3 \cdot \dfrac{8}{20+m}\left(1 - \dfrac{8}{20+m}\right) = \dfrac{24(12+m)}{(20+m)^2}$ **M1 A1**

$\dfrac{24(12+m)}{(20+m)^2} < \dfrac{1}{2} \Rightarrow m > 17.9$ (GDC) **M1 A1**

smallest $\boxed{m = 18}$ **A1**

TI-Nspire CX: Question 51

Table: ctrl T on a Graphs page with `f1(x)=...`, scroll to the first x that works. `f1(x)=24(12+x)/(20+x)^2`; first x with value < 0.5 is 18.

52 ← question *SME Very Hard · Q11*

(a) $44p = 33 \Rightarrow p = \dfrac{3}{4}$; $8 \times \dfrac{3}{4} = \boxed{6}$ **M1 A1**

(b) $8 \cdot \dfrac{3}{4} \cdot \dfrac{1}{4} = \boxed{\dfrac{3}{2}}$ **M1 A1**

(c) $\binom{8}{4}\left(\dfrac{3}{4}\right)^4\left(\dfrac{1}{4}\right)^4 = 70 \cdot \dfrac{81}{256} \cdot \dfrac{1}{256} = \boxed{\dfrac{2835}{32768}}$ **M1 A1 M1 A1**

53 answer → No calculator **SME Very Hard · Q12**

[Maximum mark: 6]

A box contains many marbles, $\frac{2}{5}$ of which are blue. The rest are white. Marbles are picked at random from the box. From drawing three marbles, with replacement, the probability of getting x blue marbles is $\frac{36}{125}$. Find the value of x.

53 ← question *SME Very Hard · Q12*

$X \sim \mathrm{B}\left(3, \frac{2}{5}\right)$: $\mathrm{P}(X = x) = \binom{3}{x}\left(\frac{2}{5}\right)^{x}\left(\frac{3}{5}\right)^{3-x}$ **M1**

$x = 0 : \frac{27}{125},\ x = 1 : \frac{54}{125},\ x = 2 : 3 \cdot \frac{4}{25} \cdot \frac{3}{5} = \frac{36}{125}$ ✓, $x = 3 : \frac{8}{125}$ **M1**

A1 A1

$\boxed{x = 2}$ **A2**

54 answer →

Calculator RV · Q20

[Maximum mark: 8]

A factory makes toys. The probability that a toy is defective is 0.08. A random sample of 50 toys is tested.

(a) Find the expected number of defective toys. [2]

(b) Find the probability that there is at least one defective toy in the sample. [3]

(c) Given that there is at least one defective toy in the sample, find the probability that there are at most three defective toys. [3]

55 answer →

Calculator RV · Q18

[Maximum mark: 8]

AB Technologies produces DC motors of which 5% are defective. The motors are packaged into boxes, each box containing 60 motors. A box is selected at random.

(a) Find the probability that the box contains: (i) exactly one defective motor; (ii) at least two defective motors. [4]

(b) Given that there are at least two defective motors in the box, find the probability that there are at most four defective motors. [2]

(c) James orders 5 boxes of motors for his company. Find the probability that James will receive at least one but no more than fifteen defective motors in his order. [2]

54 ← question *RV · Q20*

$X \sim \mathrm{B}(50, 0.08)$

(a) $\boxed{4}$ **M1 A1**

(b) $1 - \mathrm{P}(X = 0) = \boxed{0.985}$ **M1 A2**

(c) $\dfrac{\mathrm{P}(1 \le X \le 3)}{\mathrm{P}(X \ge 1)} = \dfrac{0.410}{0.985} = \boxed{0.416}$ **M1 A1 A1**

TI-Nspire CX: Question 54

`binomCdf(50,0.08,1,50); binomCdf(50,0.08,1,3)`

55 ← question *RV · Q18*

$X \sim \mathrm{B}(60, 0.05)$

(a)(i) $\boxed{0.145}$ (ii) $\mathrm{P}(X \ge 2) = \boxed{0.808}$ **A2 A2**

(b) $\dfrac{\mathrm{P}(2 \le X \le 4)}{\mathrm{P}(X \ge 2)} = \dfrac{0.628}{0.808} = \boxed{0.777}$ **M1 A1**

(c) $Y \sim \mathrm{B}(300, 0.05)$: $\mathrm{P}(1 \le Y \le 15) = \boxed{0.568}$ **M1 A1**

TI-Nspire CX: Question 55

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).

`binomPdf(60,0.05,1); binomCdf(300,0.05,1,15)`

56 answer →

Calculator **RV · Q30**

[Maximum mark: 6]

The probability of obtaining heads when a biased coin is tossed is 0.65. The coin is tossed seven times. Find the probability of obtaining:

(a) at least four heads; [2]

(b) the fourth tail is on the seventh toss. [4]

57 answer →

Calculator **SME Medium · Q6**

[Maximum mark: 5]

Giovanni is rolling a biased dice, for which the probability of landing on a two is 0.25. He rolls the dice 10 times and records the number of times that it lands on a two. Find the probability that (i) the dice lands on a two 4 times (ii) the dice lands on a two 4 times by landing on a two 3 times in the first 9 rolls, and then landing on a two on the tenth roll.

58 answer →

Calculator **Oxford Exam-style · Q19**

[Maximum mark: 7]

In a mass production of batteries, the probability that one battery is defective is 0.5%. Given that the batteries are packed randomly in packs of 10 and quality control inspect batteries from randomly selected packs, find the probability that a selected pack contains

(a) one defective battery [2]

(b) not more than one defective battery [2]

(c) exactly 1 defective battery, given that it has not more than one defective battery. [3]

56 ← question *RV · Q30*

(a) $H \sim \mathrm{B}(7, 0.65)$: $\mathrm{P}(H \geq 4) = \boxed{0.800}$ **M1 A1**

(b) exactly 3 tails in the first 6 tosses, then a tail: $T \sim \mathrm{B}(6, 0.35)$ **M1**

$\mathrm{P}(T = 3) \times 0.35 = 0.235491 \times 0.35 = \boxed{0.0824}$ **M1 A1 A1**

TI-Nspire CX: Question 56

`binomCdf(7,0.65,4,7);` menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(6,0.35,3)`

57 ← question *SME Medium · Q6*

(i) $X \sim \mathrm{B}(10, 0.25)$: $\mathrm{P}(X = 4) = \boxed{0.146}$ **M1 A1**

(ii) $Y \sim \mathrm{B}(9, 0.25)$: $\mathrm{P}(Y = 3) \times 0.25 = 0.233597 \times 0.25 = \boxed{0.0584}$ **M1 M1 A1**

TI-Nspire CX: Question 57

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(9,0.25,3)`

58 ← question *Oxford Exam-style · Q19*

$X \sim \mathrm{B}(10, 0.005)$

(a) $\mathrm{P}(X = 1) = \boxed{0.0478}$ **M1 A1**

(b) $\mathrm{P}(X \leq 1) = \boxed{0.999}$ **M1 A1**

(c) $\dfrac{\mathrm{P}(X = 1)}{\mathrm{P}(X \leq 1)} = \dfrac{0.047794}{0.998905} = \boxed{0.0478}$ **M1 M1 A1**

TI-Nspire CX: Question 58

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value). `binomPdf(10,0.005,1)`

59 answer → Calculator **Oxford Exam-style · Q23**

[Maximum mark: 9]

An exam consists of 20 true/false questions. John knows the correct answers to 8 of the questions and decides to choose at random the answers to the remaining questions.

(a) Find the probability that John answers (i) 10 questions correctly (ii) all the 20 questions correctly. [4]

(b) A student gains 2 marks for each correct answer, has one mark subtracted for each incorrect answer, and gains no marks if the question is left unanswered. Compare the expected number of marks when the student answers all the questions with the marks she will gain if she just answers the questions to which she knows the correct answer. [5]

59 ← question *Oxford Exam-style · Q23*

$X \sim \mathrm{B}(12, 0.5)$ = correct guesses

(a)(i) 10 correct $\Rightarrow X = 2$: $\boxed{0.0161}$ **M1 A1** (ii) $X = 12$: $0.5^{12} = \boxed{0.000244}$ **M1 A1**

(b) $\mathrm{E}(X) = 6$ correct, 6 wrong: $16 + 12 - 6 = 22$ marks; known only: $8 \times 2 = 16$ marks **M1 A1 A1**

Answering everything gives $\boxed{6}$ more marks on average **R1 A1**

(Check: $\mathrm{E}(X) = np = 12 \times 0.5 = 6$, not npq.)

TI-Nspire CX: Question 59

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).

```
binomPdf(12,0.5,2)
```

60 answer → Calculator **RV · Q42**

[Maximum mark: 15]

A burger shop serves veggie burgers and chicken burgers. Each customer will choose one of the following three options; buy only a veggie burger, buy only a chicken burger or buy both a veggie burger and a chicken burger. The probability that a customer buys a veggie burger is 0.65 and the probability that a customer buys a chicken burger is 0.54.

(a) Find the probability that a customer chosen at random will buy: (i) both a veggie burger and a chicken burger; (ii) only a chicken burger. [5]

(b) On a typical day, 600 customers come to the burger shop. Find: (i) the expected number of veggie burgers sold on a typical day; (ii) the probability that more than 400 veggie burgers will be sold on a typical day. [4]

(c) It is known that 44% of the customers who come to the burger shop are females, and that 75% of the female customers buy a chicken burger. (i) A customer is selected at random. Find the probability that the customer is female and buys a chicken burger. (ii) A male customer is selected at random. Find the probability that he buys a chicken burger. [6]

60 ← question *RV · Q42*

(a)(i) $1 = 0.65 + 0.54 - \mathrm{P}(V \cap C) \Rightarrow \boxed{0.19}$ **M1 A1** (ii)

$0.54 - 0.19 = \boxed{0.35}$ **M1 A2**

(b) $X \sim \mathrm{B}(600, 0.65)$: (i) $\boxed{390}$ (ii) $\mathrm{P}(X \geq 401) = \boxed{0.185}$ **A1 M1 A2**

(c)(i) $0.44 \times 0.75 = \boxed{0.33}$ **M1 A1** (ii) $\mathrm{P}(C \cap M) = 0.54 - 0.33 = 0.21$,

$\dfrac{0.21}{0.56} = \boxed{0.375}$ **M1 M1 A1 A1**

TI-Nspire CX: Question 60

`binomCdf(600,0.65,401,600)` → 0.185

61 answer → Calculator **RV · Q34**

[Maximum mark: 15]

Jane commutes to work five days a week and travels by either bike or train. When it rains, the probability Jane rides her bike is 0.2. When it does not rain, the probability she catches the train is 0.3. The probability it rains on any given day is 0.4.

(a) On a randomly selected day, find the probability Jane catches the train to work. [4]

(b) Given that Jane catches the train, find the probability it was raining that day. [3]

(c) In a randomly chosen work week, find the probability Jane catches the train on exactly two out of the five days. [3]

(d) If Jane travels to work 48 weeks a year, riding her bike is free and the train costs \$1.70 per day, calculate how much Jane would expect to pay per year on her commute to work. [5]

61 ← question *RV · Q34*

(a) $0.4(0.8)+0.6(0.3)=\boxed{0.5}$ **M1 A1 A2**

(b) $\dfrac{0.32}{0.5}=\boxed{0.64}$ **M1 A2**

(c) $X\sim \mathrm{B}(5,0.5)$: $\mathrm{P}(X=2)=\boxed{0.3125}$ **M1 A2**

(d) $48\times 5\times 0.5\times \$1.70=\$\boxed{204}$ **M1 M1 A1 A2**

TI-Nspire CX: Question 61

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).

```
binomPdf(5,0.5,2)
```

62 answer → Calculator **RV**

[Maximum mark: 15]

A random variable X has a probability distribution given in the following table.

x	0	1	2
$\mathrm{P}(X = x)$	p	$\frac{8}{15}$	q

(a) Given that $\mathrm{E}(X) = 0.8$, find the exact value of: (i) q; (ii) p. [4]

(b) A bag contains black and red balls, with at least two of each colour. Two balls are drawn from the bag without replacement. The number of red balls drawn is given by the random variable X. (i) Write down the probability of drawing two red balls. (ii) Explain why the probability of drawing two black balls is $\frac{1}{3}$. (iii) The bag contains a total of ten balls of which k are black. Find k. [5]

(c) A game is played in which two balls are drawn from the bag of ten balls, without replacement. A player wins a prize if two black balls are drawn. Olivia plays the game twelve times. Find the probability that she wins exactly three prizes. [2]

(d) William plays the game until he wins three prizes. Find the probability that he wins his third prize on his tenth attempt. [4]

62 ← question *RV*

(a)(i) $\frac{8}{15}+2q=0.8 \Rightarrow \boxed{q=\frac{2}{15}}$ **M1 A1** (ii) $p=1-\frac{8}{15}-\frac{2}{15}=\boxed{\frac{1}{3}}$ **M1 A1**

(b)(i) $\mathrm{P}(X=2)=\boxed{\frac{2}{15}}$ (ii) two black $\Leftrightarrow X=0$, so $p=\frac{1}{3}$ (iii) $\frac{k}{10}\cdot\frac{k-1}{9}=\frac{1}{3} \Rightarrow k^2-k-30=0 \Rightarrow \boxed{k=6}$ **A1 R1 M1 A1 A1**

(c) $S\sim \mathrm{B}\left(12,\frac{1}{3}\right)$: $\mathrm{P}(S=3)=\boxed{0.212}$ **M1 A1**

(d) $T\sim \mathrm{B}\left(9,\frac{1}{3}\right)$: $\mathrm{P}(T=2)\times\frac{1}{3}=0.234111\times\frac{1}{3}=\boxed{0.0780}$ **M1 M1 A1 A1**

TI-Nspire CX: Question 62

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: *n*, *p*, X value (one value). `binomPdf(12,1/3,3);` menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: *n*, *p*, X value (one value). `binomPdf(9,1/3,2)`

63 answer → Calculator **SME Very Hard · Q1**

[Maximum mark: 8]

(a) In London, England, the probability that a football player, who plays club football, is left-footed is 0.24. One particular squad, Raiders FC, has 25 players. Find the probability that Raiders FC has exactly the expected number of left-footed players. [3]

(b) Find the probability that there are no left-footed players on the Raiders FC squad. [2]

(c) Find the probability that there are more right-footed players than left-footed players on the Raiders FC squad. [2]

(d) Suggest one assumption that has been made in this question. [1]

63 ← question *SME Very Hard · Q1*

$X \sim \mathrm{B}(25, 0.24)$

(a) $\mathrm{E}(X) = 6$; $\mathrm{P}(X = 6) = \boxed{0.184}$ **M1 A1 A1**

(b) $0.76^{25} = \boxed{0.00105}$ **M1 A1**

(c) left ≤ 12: $\mathrm{P}(X \leq 12) = \boxed{0.998}$ **M1 A1**

(d) e.g. players are independent / each player is either left- or right-footed **R1**

TI-Nspire CX: Question 63

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: *n*, *p*, X value (one value).

`binomPdf(25,0.24,6); binomCdf(25,0.24,0,12)`

64 answer → Calculator **SME Very Hard · Q2**

[Maximum mark: 6]

(a) When a pizza store receives a delivery order via their app the customer is told a predicted delivery time. The probability that the pizza store delivers the order before the predicted delivery time is $\frac{9}{20}$. During a given week, the pizza store makes 160 deliveries. Find the expected number of deliveries delivered before the predicted time during that week. [2]

(b) Calculate the probability that they deliver over 100 orders before the predicted delivery time during that week. Give your answer in the form $a \times 10^k$, where $1 \leq a < 10$ and $k \in \mathbb{Z}$. [2]

(c) The manager decides that it is more important to deliver within 5 minutes of the predicted delivery time (before or after). He finds that the constant probability for this measure is $\frac{3}{4}$. Calculate the probability that over 100 orders were delivered within 5 minutes of the predicted time during the week with 160 deliveries. [2]

64 ← question *SME Very Hard · Q2*

(a) $160 \times \dfrac{9}{20} = \boxed{72}$ **M1 A1**

(b) $\mathrm{P}(X \geq 101) = \boxed{3.04 \times 10^{-6}}$ **M1 A1**

(c) $Y \sim \mathrm{B}(160, 0.75)$: $\mathrm{P}(Y \geq 101) = \boxed{0.9997}$ **M1 A1**

TI-Nspire CX: Question 64

```
binomCdf(160,0.45,101,160); binomCdf(160,0.75,101,160)
```

65 answer → Calculator **SME Very Hard · Q3**

[Maximum mark: 9]

(a) At a university, it is known from previous cohorts that the probability that a student is taller than 1.8 m is $\frac{3}{10}$ and the probability that a student weighs less than 80 kg is $\frac{9}{10}$. The university currently has 820 students. Find the mean and variance for the number of students that (i) are 1.8 m or shorter (ii) weigh 80 kg or more. [5]

(b) Assuming that a student's height and weight are independent, find the probability that 250 students or more are taller than 1.8 m and weigh less than 80 kg. [3]

(c) Comment on the assumption that a student's height and weight are independent. [1]

65 ← question *SME Very Hard · Q3*

(a)(i) $\mathrm{B}(820, 0.7)$: mean $\boxed{574}$, var $\boxed{172.2}$ (ii) $\mathrm{B}(820, 0.1)$: mean $\boxed{82}$, var $\boxed{73.8}$ **M1 A2 A2**

(b) $p = 0.3 \times 0.9 = 0.27$; $\mathrm{P}(X \geq 250) = \boxed{0.0143}$ **M1 M1 A1**

(c) unrealistic — taller students tend to weigh more **R1**

TI-Nspire CX: Question 65

```
binomCdf(820,0.27,250,820)
```

66 answer → Calculator **SME Very Hard · Q4**

[Maximum mark: 9]

(a) A swimming training session for Ben includes swimming 12 lengths of 100 m. Ben can swim 100 m in less than 60 s with a constant probability of $\frac{5}{6}$. Let X be the number of times Ben completes the 100 m length in less than 60 s during a training session. Calculate the probability that Ben completes the 100 m in less than 60 s for all the 12 lengths. [2]

(b) Find the probability that Ben completes the 100 m in less than 60 s at least twice but no more than 9 times during a training session. [3]

(c) Ben's younger brother, Zack, can complete the 100 m in less than 60 s with a constant probability of $\frac{2}{5}$. Find the probability that both Ben and Zack complete the 100 m in less than 60 s more than four times during a training session. [3]

(d) Suggest a reason as to why the probability for Ben and Zack to complete the 100 m in less than 60 s is not constant. [1]

66 ← question *SME Very Hard · Q4*

(a) $\left(\frac{5}{6}\right)^{12} = \boxed{0.112}$ **M1 A1**

(b) $\mathrm{P}(2 \leq X \leq 9) = \boxed{0.323}$ **M1 A2**

(c) $\mathrm{P}(X \geq 5) \times \mathrm{P}(Z \geq 5) = 0.999917 \times 0.561822 = \boxed{0.562}$ **M1 M1 A1**

(d) e.g. they get tired in later lengths **R1**

TI-Nspire CX: Question 66

```
binomCdf(12,5/6,2,9); binomCdf(12,0.4,5,12)
```

67 answer →

Calculator SME Very Hard · Q5

[Maximum mark: 7]

(a) A multiple-choice test consists of 50 questions. Each question has 5 options of which only one is correct. Henry randomly chooses one of the five answers for each question. Find the expected number of questions Henry answers incorrectly. [1]

(b) Henry must get at least 25 of the questions correct to pass the test. Find the probability Henry passes the test. Give your answer in the form $a \times 10^k$, where $1 \leq a < 10$ and $k \in \mathbb{Z}$. [2]

(c) As Henry is going through the test, he realizes he is certain that he knows the answer to 12 of the questions. Find the probability Henry passes, given he gets these 12 questions all correct. [4]

68 answer →

Calculator SME Very Hard · Q6

[Maximum mark: 8]

(a) A mall has a "clothing" section and an "other" section. The probability that a randomly observed customer goes to the clothing section is 0.78, and the probability that a randomly observed customer goes to the other section is 0.42. Assume that each customer goes to at least one section. Find the probability that a randomly observed customer (i) goes to both sections, (ii) only goes to the clothing section. [4]

(b) On a given day the mall has 341 customers. Find (i) the expected number of customers that go to the other section, (ii) the probability that at least 250 customers visit the clothing section. [4]

67 ← question *SME Very Hard · Q5*

(a) $50 \times 0.8 = \boxed{40}$ **A1**

(b) $X \sim \mathrm{B}(50, 0.2)$: $\mathrm{P}(X \geq 25) = \boxed{2.09 \times 10^{-6}}$ **M1 A1**

(c) needs ≥ 13 of the other 38: $Y \sim \mathrm{B}(38, 0.2)$, $\mathrm{P}(Y \geq 13) = \boxed{0.0288}$

M1 M1 M1 A1

TI-Nspire CX: Question 67

```
binomCdf(50,0.2,25,50); binomCdf(38,0.2,13,38)
```

68 ← question *SME Very Hard · Q6*

(a)(i) $0.78 + 0.42 - 1 = \boxed{0.2}$ (ii) $0.78 - 0.2 = \boxed{0.58}$ **M1 A1 M1 A1**

(b)(i) $341 \times 0.42 = \boxed{143.22}$ (not rounded) **M1 A1** (ii)

$\mathrm{P}(X \geq 250) = \boxed{0.983}$ **M1 A1**

TI-Nspire CX: Question 68

```
binomCdf(341,0.78,250,341)
```

69 answer → Calculator **SME Very Hard · Q7**

[Maximum mark: 9]

(a) 360 students at a school are divided into three groups. The probability of a student being put into group A is $\frac{1}{12}$. A student is twice as likely to be put into group B than group C. Find the expected number of students put into each group. [4]

(b) 20 students are chosen at random to form a sample. Find the probability that at least four of the students are in group A. [2]

(c) Find the probability that less than 15 of the students are in groups A or B. [2]

(d) The headteacher now wants exactly $\frac{1}{12}$ of the students to be in group A. Explain why the number of group A students in the sample can not be modelled by a binomial distribution. [1]

69 ← question *SME Very Hard · Q7*

(a) $\mathrm{P}(C) = \dfrac{11}{36}$, $\mathrm{P}(B) = \dfrac{22}{36}$: A $\boxed{30}$, B $\boxed{220}$, C $\boxed{110}$ **M1 A1 A1 A1**

(b) $\mathrm{B}(20, \dfrac{1}{12})$: $\mathrm{P}(X \geq 4) = \boxed{0.0796}$ **M1 A1**

(c) $p = \dfrac{25}{36}$: $\mathrm{P}(X \leq 14) = \boxed{0.605}$ **M1 A1**

(d) group A is then a fixed 30 of 360 students; sampling without replacement changes p (not independent) **R1**

TI-Nspire CX: Question 69

```
binomCdf(20,1/12,4,20); binomCdf(20,25/36,0,14)
```

70 answer →

Calculator **SME Very Hard · Q8**

[Maximum mark: 9]

(a) A bag of 500 marbles are divided into five colours: red, orange, yellow, green and blue. There are x red marbles, $3x$ orange, $5x$ yellow, $7x$ green and $9x$ blue marbles. Find the value of x. [2]

(b) Maeve selects a marble at random, records its colour and then returns it to the bag. Maeve follows this process 56 times. Find the probability that none of the selected marbles are red. [2]

(c) Find the probability that at least a quarter of the selected marbles are red or blue. [4]

(d) Maeve decides not to return the marbles to the bag after she records their colours. State, with a reason, whether this would change the answers to parts (b) and (c). [1]

71 answer →

Calculator **SME Very Hard · Q9**

[Maximum mark: 7]

(a) In a restaurant, it is known that there is an 8 out of 15 chance that a guest will ask for water with their meal. A random sample of 25 guests are sampled. Find the probability that (i) over half of the guests ask for water, (ii) between 24% and 92%, inclusive, of the guests ask for water. [4]

(b) A second random sample of 100 guests is selected. Let X be the number of guests that ask for water. Find the smallest value of n such that $P(X \leq n) \geq 0.95$. [3]

70 ← question *SME Very Hard · Q8*

(a) $25x = 500 \Rightarrow \boxed{x = 20}$ **M1 A1**

(b) $p = \dfrac{20}{500} = 0.04$; $0.96^{56} = \boxed{0.102}$ **M1 A1**

(c) $p = \dfrac{200}{500} = 0.4$; $\mathrm{P}(X \geq 14) = \boxed{0.994}$ **M1 M1 M1 A1**

(d) yes — without replacement the trials are not independent, p changes **R1**

TI-Nspire CX: Question 70

`binomCdf(56,0.4,14,56)`

71 ← question *SME Very Hard · Q9*

(a) $\mathrm{B}(25, \dfrac{8}{15})$: (i) $\mathrm{P}(X \geq 13) = \boxed{0.632}$ (ii) $6 \leq X \leq 23$: $\boxed{0.999}$ **M1 A1 M1 A1**

(b) $\mathrm{B}(100, \dfrac{8}{15})$: $\mathrm{P}(X \leq 61) = 0.946$, $\mathrm{P}(X \leq 62) = 0.963$: $\boxed{n = 62}$ **M1 A1 A1**

TI-Nspire CX: Question 71

`binomCdf(100,8/15,0,n)` for $n = 61, 62$ (or a table).

72 answer → Calculator **SME Very Hard · Q10**

[Maximum mark: 6]

(a) In a game, players select pieces of string from a box containing a large number of pieces of string of different lengths. The length, in cm, of a randomly chosen piece of string has uniform distribution over the interval $[4, 9]$. A piece of string is selected at random. Find the probability that the piece of string is shorter than 7.5 cm. [2]

(b) If a player selects five pieces of string, then they win a chocolate bar if the length of the longest piece of string is more than 7.5 cm. A player selects five pieces of string; find the probability of winning a chocolate bar. [2]

(c) If a player selects eight pieces of string, they win a box of chocolates if five or more of the pieces of string are shorter than 5.2 cm. Find the probability of winning a box of chocolates. [2]

72 ← question *SME Very Hard · Q10*

(a) $\dfrac{7.5-4}{9-4} = \boxed{0.7}$ **M1 A1**

(b) $1-0.7^5 = \boxed{0.832}$ **M1 A1**

(c) $p = \dfrac{1.2}{5} = 0.24$; $\mathrm{B}(8, 0.24)$: $\mathrm{P}(X \geq 5) = \boxed{0.0230}$ **M1 A1**

TI-Nspire CX: Question 72

```
binomCdf(8,0.24,5,8)
```

73 answer → Calculator **RV · Q54**

[Maximum mark: 21]

Ten balls numbered $2, 2, 3, 3, 3, 5, 5, 5, 5, 5$ are placed in a bag. Balls are taken one at a time from the bag at random and the number noted. Throughout the question a ball is always replaced before the new ball is taken.

(a) A single ball is taken from the bag. Let X denote the value shown on the ball. Find $\mathrm{E}[X]$. [2]

(b) Three balls are taken from the bag. Find the probability that: (i) the sum of the three numbers is 9; (ii) the median of the three numbers is 3. [6]

(c) Twelve balls are taken from the bag. Find the probability that more than five of the balls are numbered 2. [3]

(d) Determine the least number of balls that must be taken from the bag for the probability of taking out at least one ball numbered 5 to be greater than 0.9. [3]

(e) Another bag also contains balls numbered 2, 3 or 5. Nine balls are to be taken from this bag at random. It is calculated that the expected number of balls numbered 2 is 4.5, and the variance of the number of balls numbered 3 is 1.25. Find the least possible number of balls numbered 5 in this bag. [7]

73 ← question *RV · Q54*

(a) $2(0.2)+3(0.3)+5(0.5)=\boxed{3.8}$ **M1 A1**

(b)(i) $\{2,2,5\}$ (3 ways) or $\{3,3,3\}$: $3(0.2)^2(0.5)+0.3^3=\boxed{0.087}$ **M1 M1 A1**

(ii) $\{2,3,3\},\{3,3,3\},\{3,3,5\},\{2,3,5\}$: $0.054+0.027+0.135+0.18=\boxed{0.396}$ **M1 M1 A1**

(c) $\mathrm{B}(12,0.2)$: $\mathrm{P}(Y\geq 6)=\boxed{0.0194}$ **M1 A2**

(d) $0.5^N<0.1\Rightarrow N>3.32$: $\boxed{4}$ balls **M1 A1 A1**

(e) $9p=4.5\Rightarrow p=\dfrac{1}{2}$; $9q(1-q)=1.25\Rightarrow q=\dfrac{1}{6}$ (reject $\dfrac{5}{6}$) **M1 A1 M1 A1 R1**

$\mathrm{P}(5)=\dfrac{1}{3}$; smallest bag 6 balls: $\boxed{2}$ fives **A1 A1**

TI-Nspire CX: Question 73

```
binomCdf(12,0.2,6,12); solve(9q(1-q)=1.25,q)
```

74 answer → Calculator **RV · Q53**

[Maximum mark: 22]

Ten balls numbered $1, 2, 2, 3, 3, 3, 4, 4, 4, 4$ are placed in a bag. Balls are taken one at a time from the bag at random and the number noted. Throughout the question a ball is always replaced before the new ball is taken.

(a) A single ball is taken from the bag. Let X denote the value shown on the ball. Find $\mathrm{E}[X]$. [2]

(b) Three balls are taken from the bag. Find the probability that: (i) the sum of the three numbers is 8; (ii) the median of the three numbers is 4. [6]

(c) Fifteen balls are taken from the bag. Find the probability that less than six of the balls are numbered 3. [3]

(d) Find the least number of balls that must be taken from the bag for the probability of taking out at least one ball numbered 4 to be greater than 0.95. [3]

(e) Another bag also contains balls numbered 1, 2, 3 or 4. Twenty balls are to be taken from this bag at random. It is calculated that the expected number of balls numbered 2 is 2.5, the expected number of balls numbered 3 is 5, and the variance of the number of balls numbered 1 is 1.8. Find the least possible number of balls numbered 4 in this bag. [8]

74 ← question *RV · Q53*

(a) $\dfrac{1+4+9+16}{10} = \boxed{3}$ **M1 A1**

(b)(i) $\{1,3,4\}$ (6), $\{2,2,4\}$ (3), $\{2,3,3\}$ (3): $0.072+0.048+0.054 = \boxed{0.174}$ **M1 M1 A1**

(ii) at least two 4s: $3(0.1+0.2+0.3)(0.4)^2+0.4^3 = \boxed{0.352}$ **M1 M1 A1**

(c) $\mathrm{B}(15,0.3)$: $\mathrm{P}(Y \le 5) = \boxed{0.722}$ **M1 A2**

(d) $0.6^N < 0.05 \Rightarrow N > 5.86$: $\boxed{6}$ **M1 A1 A1**

(e) $q = \dfrac{1}{8}$, $r = \dfrac{1}{4}$, $20p(1-p) = 1.8 \Rightarrow p = \dfrac{1}{10}$ (reject $\dfrac{9}{10}$) **M1 A1 A1 M1 A1 R1**

$\mathrm{P}(4) = 1 - \dfrac{19}{40} = \dfrac{21}{40}$; smallest bag 40: $\boxed{21}$ **A1 A1**

TI-Nspire CX: Question 74

```
binomCdf(15,0.3,0,5); solve(20p(1-p)=1.8,p)
```